(c) Tamil Nadu's wind power capacity was about 24% of the country in the year 2021.
(d) Madhya Pradesh has third rank in the country in wind power capacity.

M.P. P.C.S. (Pre) 2022

Ans. (d)

Madhya Pradesh has seventh rank in the country in wind power installed capacity. As per the data of 2021, other three statements were correct. However, Gujarat has surpassed Tamil Nadu in total wind power installed capacity in 2023. As on 28 February, 2025, total wind power installed capacity in India is 48588.56 MW in which Gujarat has around 25.9% share (12583.88 MW) while Tamil Nadu has around 23.7% share (11514.64 MW).

Conductivity

Notes

Electric Conductivity :

- Within the atom, electrons nearest to the nucleus are strongly bound to positive ions (protons) of the nucleus by attractive force.
- The electrons far from the nucleus have poor attractive force.
- Due to poor attractive force, such electrons are easily removed from their original state.
- Such electrons are known as **free electrons** or **conduction electrons**.
- The electrons which are not bound to the nucleus of an atom and free to move when external energy is applied are called free electrons.
- In General Physics, any electron that is not attached to an ion, atom or molecule and is free to move under the influence of an applied electric or magnetic field is called free electron.
- A metal with a good number of a free electrons is a good conductor of electricity.
- Silver is the best conductor of electricity.
- Other metals - copper, gold and aluminium are respectively good conductors of electricity.
- Electric conductivity is also found in some liquids and gases.
- In metals, electric conductivity is due to the movement of free electrons, while in liquid & gases it is due to the movement of positive and negative ions.
- In gases, electric conductivity takes place only at a definite pressure range (approx. 10 mm Hg to 10^{-3} mm Hg). Above the maximum limit of pressure (10mm Hg) and below the minimum limit (10^{-3} mm Hg) of pressure, gases are a bad conductor of electricity.

Non-conductor :

- The matter having very low or zero numbers of free electrons are known as non-conductor of electricity.
- Such matter is also known as an **insulator**.
- The flow of charge in such matters is not possible due to the absence of free electrons.

Semiconductor :

- The matter whose electric conductivity lies between conductor and non-conductor (insulator) is known as semiconductor.
- The conductivity of such matter is due to the addition of an impurity or due to temperature effect.
- The process of adding controlled impurities to a semi-conductor is known as doping. Doped Semi- conductors are referred to as extrinsic.
- Their resistance decreases as their temperature increases, which is a behaviour opposite to that of a metal.
- Examples of semiconductors are - Germanium, Selenium, Silicon and Carbon.

Superconductivity :

- It is a phenomenon of exactly zero electrical resistance and expulsion of magnetic flux fields occurring in certain materials called superconductors, when cooled below a characteristic critical temperature.
- The temperature at which resistance of a matter suddenly becomes zero is known as 'Transition Temperature'.
- Following matters show the property of super- conductivity.

S.No.	Name of Matter	Transition Temperature (T_c)
i.	Mercury	4.2 K
ii.	Tungsten	0.01 K
iii.	Cadmium	0.56 K
iv.	Aluminium	1.19 K
v.	Stanus (Tin)	3.7 K
vi.	Lead	7.2 K

- When superconductivity was discovered by Dutch physicist Heike Onnes in 1911, it was found only at temperatures close to absolute zero (– 273.15°C). But since then, researchers have steadily uncovered materials that superconduct at higher temperatures. In recent years, scientists have accelerated that progress by focusing on hydrogen-rich materials at high pressure.
- In 2019 it was discovered that lanthanum decahydride (LaH_{10}) becomes a superconductor at around 250-260 K under a pressure of around 170 gigapascals.
- In October 2020, a near room-temperature superconductor (around 15°C or 288 K) made from hydrogen, carbon and sulphur (Carbonaceous sulphur hydride) under pressure of around 270 gigapascals was described in a paper in research magazine *Nature*. This is currently the highest temperature at which any material has shown superconductivity.

Application of Superconductivity :

a. Powerful superconducting electromagnet used in maglev trains.
b. Magnetic resonance imaging (MRI).
c. Nuclear magnetic resonance (NMR) machines.
d. Magnetic confinement fusion reactors (e.g. tokamaks).
e. Beam steering and focusing magnets used in particle accelerators.
f. Low loss of power cables.

Maglev :

- Maglev is a short form of 'Magnetic Levitation' in which trains float on a guideway using the principle of magnetic repulsion.

Question Bank

1. The substance which has a large number of free electrons and offer a low resistance is called :

(a) Insulator (b) Inductor
(c) Conductor (d) Semi-conductor

Uttarakhand P.C.S. (Pre) 2021

Ans. (c)

A conductor, or electrical conductor, is a substance or material that allows electricity to flow through it. In conductive materials, the outer electrons in each atom can easily come or go and are called free electrons. They move easily from atom to atom when voltage is applied. Conductors have a large number of free electrons with low resistance and high thermal conductivity.

2. Which elements is the best conductor of electricity?

(a) Silver (b) Copper
(c) Aluminium (d) Iron

Chhattisgarh P.C.S. (Pre) 2011
M.P.P.C.S. (Pre) 1990

Ans. (a)

Electrical conductivity is a measure of the amount of electric current which a material can carry. The most electrically conductive element is silver followed by copper, aluminium and iron.

3. Which one of the following is the best conductor of electricity?

(a) Mica (b) Copper
(c) Gold (d) Silver

U.P.P.C.S. (Mains) 2015

Ans. (d)

See the explanation of above question.

4. Which of the following is the best conductor of electricity?

(a) Aluminium (b) Copper
(c) Silver (d) Gold

U.P.P.C.S. (Mains) 2012
R.A.S./R.T.S.(Pre) 2012

Ans. (c)

See the explanation of above question.

5. Which of the following liquids is a bad conductor of electricity?

(a) Salted water (b) Orange juice
(c) Lemon juice (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (d)

A salt water is completely ionized in water. Hence, it is a good conductor of electricity. Orange and Lemon juice contain various salts and acids and are therefore good conductors of electricity. Thus, option (d) is the correct answer.

6. Assertion (A) : Copper rods are generally preferred to iron rods for making lightning conductors.

Reason (R) : Copper is better conductor of electricity than iron and is not easily oxidized under atmospheric conditions.

Code :

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.R.O./A.R.O. (Mains) 2013

Ans. (a)

Copper rods are preferred for making lightning conductor because :

(i) Copper is a better conductor of electricity than iron.
(ii) Copper is not easily oxidized in the presence of oxygen and water molecules while iron gets rusted.
(iii) Loss of energy is much less with a copper rod than with iron rod, as copper is a bad conductor of heat than iron.

7. Due to temperature variation along a conductor, potential variation occurs along it. This phenomenon is known as :

(a) Thomson effect (b) Joule effect
(c) Seebeck effect (d) Peltier effect
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Due to the temperature difference between the two points of the same conductor, there is a difference in the electron density. The electron density is higher at low temperatures than at higher temperatures. Thus, there will be a potential difference between two points of the same conductor. Therefore, when a current (charge) flows in the conductor, work is done against or along the direction of potential difference due to the flow of current. As a result of this heat is absorbed or evolved. This effect is called the Thomson effect.

8. Given below are two statements, in which one is labelled as Assertion (A) and the other as Reason (R).
Assertion (A) : A person travelling in a car with closed glass windows and doors is safe from the lightening strike.
Reason (R) : Actually a closed car behaves like a hollow conductor, so charges cannot enter inside the car.
Select the correct answer using the code given below-
Code :

(a) (A) is true but (R) is false.
(b) Both (A) and (R) are true and (R) is correct explanation of (A).
(c) (A) is false but (R) is true.
(d) Both (A) and (R) are true but (R) is not correct explanation of (A).

U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

A person travelling in a car with closed glass windows and doors is safe from the lightening strike, because closed car behaves like a closed hollow conductor (as a Faraday cage) and electric field or charge inside a closed conductor is always zero and hence, charges cannot enter inside the car.

9. Which of the following diode is used in ultra-high speed switching electronic circuits?

(a) Zener diode (b) Varactor diode
(c) Tunnel diode (d) Schottky diode

M.P. P.C.S. (Pre) 2022

Ans. (c)

Tunnel diode is used in ultra-high speed switching electronic circuits. A tunnel diode is a type of semiconductor diode that possesses a high-speed operation capability due to quantum tunneling, making it ideal for ultra-high speed switching circuits. It has effectively 'negative resistance' due to the quantum mechanical effect called tunneling.

10. The characteristic of superconductor is –

(a) High permeability (b) Low permeability
(c) Zero permeability (d) Infinite permeability

39th B.P.S.C. (Pre) 1994

Ans. (c)

In electromagnetism, permeability is a property of a material that describes the ability of a material to support the formation of a magnetic field within itself or we can say that it is the degree of magnetization that a material obtains in response to an applied magnetic field. It is represented by Greek letter μ (Mu). The permeability and electrical resistance of the superconductor is zero.

11. At which temperature superconductivity can be of tremendous economic interest saving billions of rupees?

(a) At very low temperature
(b) At a temperature when semiconductor becomes a superconductor
(c) at room temperature
(d) At a very high temperature

U.P.P.C.S. (Pre) 2000

Ans. (c)

If we could achieve superconductivity at room temperature then we can save billions of rupee through electric transmission without any power loss.

12. The highest temperature attained by a superconductor is :

(a) 24 K (b) 133 K
(c) 150K (d) 300K

U.P. Lower Sub. (Pre) 2013

Ans. (*)

When superconductivity was discovered in 1911, it was found only at temperatures close to absolute zero (– 273.15°C). But since then, researchers have steadily uncovered materials that superconduct at higher temperatures. In recent years, scientists have accelerated that progress by focusing on hydrogen-rich materials at high pressure. In 2019 it was discovered that lanthanum decahydride (LaH_{10}) becomes a superconductor at around 250-260 K under a pressure of around 170 gigapascals. In 2020, a near room-temperature superconductor (around 15°C or 288 K) made from hydrogen, carbon and sulphur under pressures of around 270 gigapascals was described in a paper in research magazine *Nature* . This is currently the highest temperature at which any material has shown superconductivity.

13. The newly discovered high temperature super conductors are –

(a) Metal alloys (b) Pure rare earth metals
(c) Ceramic oxides (d) Inorganic polymers

Uttarakhand P.C.S. (Pre) 2006
U.P.P.C.S. (Pre) 2000

Ans. (c)

Ceramic oxides play a vital role in the field of research and discovery towards superconductivity. Ceramic super conductive materials contain Mercury (Hg)/Thallium (Tl), Barium (Ba), Calcium (Ca) and Copper oxide (CO). Its discovery was first reported in 1993 and its transition temperature (T_c) was between 94 K – 135 K.

14. The conductivity of a semiconductor at zero Kelvin is–

(a) 10^5 ohm (b) 10^{-1} ohm
(c) 10^{-5} ohm (d) zero

R.A.S./R.T.S. (Pre) 1999-2000

Ans. (d)

The materials [Germanium, Silicon] whose electrical conductivity varies between conductors and dielectric are called semiconductors. At absolute zero temperature, a semiconductor behaves like a perfect dielectric.

15. The resistance of a semiconductor on heating :

(a) Remains constant (b) Decreases
(c) Increases (d) None of the above

U.P.P.C.S. (Mains) 2015

Ans. (b)

With the increase in temperature, the conductivity of the semiconductor material increases. As with increase in temperature, outermost electrons acquire energy and hence by acquiring energy, the outermost electrons leave the shell of the atom. Hence, with an increase in temperature, number of carriers in the semiconductor material increases which leads to increase in the conductivity of the material. So we can say that the semiconductor material has negative temperature coefficient i.e. with an increase in temperature, resistance decreases.

16. On heating, the resistance of a semiconductor :

(a) increases
(b) decreases
(c) remains same
(d) first increases and then decreases
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

See the explanation of above question.

17. Electric conduction in a semiconductor takes place due to-

(a) Electrons only
(b) Holes only
(c) Both electrons and holes
(d) Neither electrons nor holes

Uttarakhand P.C.S. (Pre) 2016

Ans. (c)

When electric field is applied across a semiconductor, the current conduction takes place by free electrons and holes. The free electrons are produced due to the breaking up of some covalent bonds by thermal energy. At the same times holes are created in the covalent bonds. Under the influence of electric field, conduction takes place by these free electrons and holes.

18. Which of the following technique cannot be used for generating electron-hole pairs in electronic devices?

(a) Thermal excitation (b) Impact ionization
(c) Photo excitation (d) Impurity injection

M.P. P.C.S. (Pre) 2022

Ans. (d)

Generation of electron-hole pairs in semiconductors in electronic devices can be achieved by thermal excitation (by increasing temperature), impact ionization (by which one energetic charge carrier can lose energy by the creation of other charge carriers) or photo excitation (by utilizing the photoelectric effect). Impurity injection is done for doping of semiconductors for the purpose of modulating its electrical, optical and structural properties. This process doesn't generate electron-hole pairs.

19. At absolute zero temperature, the electric resistance in semiconductor is–

(a) Infinite (b) Meager
(c) High (d) Zero

R.A.S./R.T.S. (Pre) 1992

Ans. (a)

The electrical conductivity of a semiconductor at absolute zero temperature is zero and they behave like an insulator. At this temperature their electric resistivity becomes infinite.

20. Which one of the following metals is used as semiconductor in transistors?

(a) Copper (b) Germanium
(c) Graphite (d) Silver

44th B.P.S.C. (Pre) 2000

Ans. (b)

Germanium and silicon are the main elements which are used as a semiconductor in transistors. Their conductivity on the normal/room temperature lies between the conductivity of conductors and insulators.

21. The most commonly used material for making transistors is –

(a) Aluminium (b) Silicon
(c) Copper (d) Silver

U.P.P.C.S. (Pre) 2015

Ans. (b)